Identifiability of species interactions in a living material: Bayesian calibration of an extended Hamilton principle multiphase model

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Abstract

Biofilms are living materials whose composition, and with it their mechanical response, depends on species interactions that cannot be measured directly. We calibrate a thermodynamically consistent multiphase model of a five-species oral biofilm, derived from the extended Hamilton principle, against in-vitro time courses of a commensal and a dysbiotic community, each under static and flow-reactor cultivation. All 15 entries of the symmetric interaction matrix are inferred by transitional Markov chain Monte Carlo, with the stiff forward model compiled in JAX and evaluated in parallel on a GPU (about 150 times faster than a CPU). Data enter in stages, from species composition to community viability, and each stage is checked for seed agreement, mixing and posterior mass at the prior bounds. Root-mean-square errors of the viable species fractions are 0.06–0.09 in the four conditions; under commensal static cultivation, composition and measured live fraction cannot both be matched. The data constrain only part of the interaction matrix: five and six entries in the two commensal conditions, all among the early colonisers and with the same signs where both determine them, and nine and eight in the dysbiotic conditions. The coefficient between *F. nucleatum* and *P. gingivalis* is constrained in both dysbiotic conditions but not sign-determined. The pH predicted from the calibrated composition follows the measured rise in the dysbiotic flow reactor, and removing *F. nucleatum* in silico reduces the late increase of *P. gingivalis* about fourfold, a directly testable prediction.

Keywords: Inverse characterisation, Bayesian calibration, Uncertainty quantification, Extended Hamilton principle, Oral biofilm, GPU

1. Introduction

Biofilms are soft, living materials: microbial cells embedded in a self-produced matrix whose composition evolves in time and determines their mechanical response [1]. On dental implants, a shift in the species composition of the biofilm from a commensal towards a dysbiotic community is associated with peri-implantitis, and in-vitro models of peri-implant biofilms reproduce such shifts [2]; in the ecological view of oral disease, the shift of the community, rather than a single pathogen, drives disease [3]. Early biofilms on implant abutments in vivo already contain disease-associated species [4].

Which interactions between the species drive such a shift is difficult to measure: co-culture experiments report how the composition changes over time, not the interactions themselves.

Continuum mechanics offers a route from composition to interaction. Based on the extended Hamilton principle [5], Klempt et al. [6] derived a multiphase model in which species volume fractions and viabilities evolve under a free energy that contains an interaction matrix $\mathbf{A}$, taken symmetric so that the interactions derive from this potential; thermodynamic consistency follows from the variational structure. Calibrating $\mathbf{A}$ against data is an inverse characterisation problem: the physics fixes the structure of the model, and the data determine its coefficients. Once calibrated, the model can serve as an in-silico test bed, for example for removing one species. Bayesian updating has been used to calibrate constitutive parameters of biofilm models with surrogates [7], but the interaction matrix poses two specific difficulties. First, the data give 20 independent composition values and five community viabilities for 15 unknowns, so identifiability cannot be assumed. Second, the coupled volume-fraction and viability dynamics produce posteriors that are multimodal and bounded by the prior box, for which the convergence of standard samplers is hard to verify.

Here we infer all 15 entries of $\mathbf{A}$ from the time courses of Heine et al. [2] for five oral taxa—*Streptococcus oralis* (So), *Actinomyces naeslundii* (An), *Veillonella* spp. (Vei), *Fusobacterium nucleatum* (Fn) and *Porphyromonas gingivalis* (Pg), numbered 1 to 5 in this order—in four conditions: a commensal or a dysbiotic community, each under static cultivation in well plates or dynamic cultivation in a flow reactor (HOBIC, Hannoverian Oral multispecies Biofilm Implant flow Chamber), abbreviated CS, CH, DS and DH. We combine three elements: first, transitional Markov chain Monte Carlo (TMCMC) [8, 9] with the forward model compiled in JAX [10] and evaluated for all particles in parallel on a GPU; second, a staged introduction of the data, from species composition to community viability; and third, explicit acceptance criteria for every stage, including a rule for widening prior bounds, and an analysis of which interactions the data determine. The data determine only a subset of $\mathbf{A}$, which we identify entry by entry; where the commensal conditions overlap, the identified entries agree in sign across cultivation modes, whereas the Vei–Pg coefficient of the dysbiotic community is negative in the flow reactor but close to zero under static cultivation.

2. Model

At a material point, species $i = 1, \dots, 5$ (So, An, Vei, Fn, Pg) occupies the volume fraction $\phi_i \in (0,1)$, of which the fraction $\psi_i \in (0,1)$ is alive; the living volume fraction is $\bar{\phi}_i = \phi_i \psi_i$. A void fraction ϕ_0 closes the volume, $\sum_{l=0}^{5} \phi_l = 1$. As in Klempt et al. [6], the evolution equations result from the stationarity of the rate functional (derivation in Supplementary Section S1)

$$\dot{\Psi} + \Delta + C \to \operatorname{stat}_{(\dot{\phi}, \dot{\psi}, \gamma)}, \tag{1}$$

the local form of the extended Hamilton principle [5, 11, 12, 13], with the free energy, dissipation and constraint

$$\Psi = -\tfrac{1}{2}\, c^*\, \bar{\boldsymbol{\phi}}^{\mathsf{T}} \mathbf{A}\, \bar{\boldsymbol{\phi}}, \qquad \Delta = \tfrac{1}{2}\, \dot{\bar{\boldsymbol{\phi}}}^{\mathsf{T}} \boldsymbol{\eta}\, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2}\, \dot{\boldsymbol{\phi}}^{\mathsf{T}} \boldsymbol{\eta}\, \dot{\boldsymbol{\phi}}, \qquad C = \gamma\Big(\sum_{l=0}^{5} \phi_l - 1\Big). \tag{2}$$

Here c^* is the nutrient concentration, $\boldsymbol{\eta} = \operatorname{diag}(\eta_i)$ with $\eta_i = 1$, and γ is a Lagrange multiplier; the antibiotic term of the original model vanishes because the experiments contain no antibiotic. For $n = 5$ species, stationarity gives $2n + 2 = 12$ coupled non-linear equations, for $i = 1, \dots, 5$,

$$0 = -c^* \psi_i (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \eta_i\Big(\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i\Big) + \gamma, \tag{3}$$

$$0 = -c^* \phi_i (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \eta_i\Big(\dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i\Big) + \gamma, \tag{4}$$

together with $0 = \gamma + \dot{\phi}_0$ and the volume constraint. They are integrated by an implicit Euler scheme with Newton iterations (2500 steps covering the 21-day experiment), with a barrier term that diverges at 0 and 1 keeping ϕ_i, ψ_i inside $(0,1)$. Only the symmetric part of $\mathbf{A}$ enters the quadratic free energy, so $\mathbf{A}$ is taken symmetric, with 15 independent entries, which form the parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{15}$. The coefficient a_{ij} is an effective, undirected interaction of the model; it lumps mechanisms that the model does not resolve, such as metabolite exchange or co-aggregation. Only the product $c^* a_{ij}$ enters, so we fix $c^* = 25$ to keep $a_{ij} = \mathcal{O}(1)$. Removing a species is represented by holding its volume fraction at zero.

3. Bayesian inference

Likelihood.. The composition data are the species fractions of viable cells at days 3, 6, 10, 15 and 21 (day 1 is the initial condition), measured by quantitative real-time PCR after propidium monoazide (PMA) treatment [2], with the total-count fractions without PMA measured on the same samples (their Figs. 3A and S1). We use the measured replicate values provided by the authors (three to five biological replicates per time point, each in technical triplicate) and the medians over them (values read off the published box plots deviate by up to 0.42; Supplementary Section S11). Model time is $t = s\,d$ ($s = 0.2375/21$) with the day-1 composition as the initial state; starting at day 1 instead ($t = s\,(d-1)$), as the medium change after 24 h in the flow reactor [2] would suggest, changes the composition errors of the maximum a posteriori estimate (MAP) by at most 0.014 (0.009, 0.014, 0.010 and 0.014 in CS, CH, DS and DH; in DH the viable-composition error decreases). The two compositions differ (root-mean-square error, RMSE, between their medians 0.101, 0.087, 0.119 and 0.087 in CS, CH, DS and DH; Supplementary Section S12) because the species differ in viability, so each is compared with its own model counterpart. The viable fractions y^{v}_{ik} at time t_k are compared with the living volume fractions $\hat{y}^{\mathrm{v}}_{ik} = \phi_i\psi_i / \sum_j \phi_j\psi_j$, the total-count fractions y^{t}_{ik} with $\hat{y}^{\mathrm{t}}_{ik} = \phi_i / \sum_j \phi_j$, and the community live fraction ℓ_k, the fraction of membrane-intact cells from SYTO9/propidium iodide (LIVE/DEAD BacLight) staining (image based, not resolved by species; Supplementary Section S11), with $\hat{\ell}_k = \sum_i \phi_i\psi_i / \sum_i \phi_i$, through the weighted Gaussian log-likelihood

$$\ln p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta}) = -\frac{1}{2}\sum_k \sum_{i=1}^{5} w_{ik} \left[\frac{\left(y^{\mathrm{v}}_{ik} - \hat{y}^{\mathrm{v}}_{ik}\right)^2}{(\sigma^{\mathrm{v}}_{ik})^2} + \frac{\left(y^{\mathrm{t}}_{ik} - \hat{y}^{\mathrm{t}}_{ik}\right)^2}{(\sigma^{\mathrm{t}}_{ik})^2} \right] - \frac{\lambda_\psi}{2}\sum_k \frac{(\ell_k - \hat{\ell}_k)^2}{\sigma_\ell^2} + \text{const}, \tag{5}$$

all evaluated at t_k for parameters $\boldsymbol{\theta}$, with σ^{v}_{ik} and σ^{t}_{ik} the standard deviations (SD) of species i at time t_k estimated from the spread of the measured replicates (interquartile range/1.35 over all measurements, at least 0.05) and σ_ℓ the SD of the live fraction between biofilms (five images per biofilm, three to nine biofilms per time point), averaged over the time points. The weights w_{ik} act on both compositions alike: they up-weight Pg ($\lambda_{\mathrm{Pg}} = 5$), whose late increase is the main dynamic feature of the dysbiotic data, and the last two time points ($\lambda_{\mathrm{late}} = 3$), and down-weight species with a mean fraction below 5% ($\lambda_{\mathrm{rare}} = 0.1$; Supplementary Table S2); they change the error of each channel by at most 0.04, and without them the late Pg increase in DH is overestimated (day-21/day-15 ratio 24 against 14.9 measured; Supplementary Section S10). The live fraction is weighted by $\lambda_\psi = 1.5$ in the commensal conditions, 2.0 in DS and 3.0 in DH (Supplementary Table S2). In stages (i) and (ii) (below) the viabilities are not evolved but fixed to the values implied by the two compositions and the live fraction, $\psi_i(t_k) = (y^{\mathrm{v}}_{ik}/y^{\mathrm{t}}_{ik})\,\ell_k$ clipped to $[0.01, 0.99]$ (the community value ℓ_k for species below 2% of the total count; Supplementary Section S13), and the live-fraction term is omitted; in stages (iii) and (iv) ψ_i follows Eq. (4) and all three terms are used. The measured pH under dynamic cultivation is not used in the inference and serves as a check (Section 4).

Prior and stages.. The prior is uniform on a box $[l_k, u_k]$ for each entry. The stage-(i) boxes were set from earlier calibrations of a previous version of the model on the same data, and some of them exclude negative values or allow only small ones (CS: $a_{11} \in [0, 2]$; DH: $a_{11}, a_{22}, a_{44} \in [0, 5]$, $a_{12} \in [0.5, 3]$, $a_{23} \in [0, 3]$, $a_{45} \in [-0.5, 6]$), so the signs of these entries are partly set by the prior in stages (i)–(ii); all of these bounds were widened in stage (iii) (Supplementary Table S4). Inference proceeds in four stages on the same data (Fig. 1): (i) composition likelihoods with ψ_i fixed to the data, on the box; (ii) as (i), on the box narrowed to the stage-(i) posterior mean $\pm 4\,\mathrm{SD}$; (iii) full likelihood with ψ_i evolved, back on the full box, half of the initial particles drawn around the stage-(ii) MAP and half from the prior; (iv) as (iii), on the box narrowed to the stage-(iii) posterior mean $\pm 4\,\mathrm{SD}$. A CH stage-(iii) run started entirely from the prior reaches the same maximum log-likelihood and the same posterior of a_{33} and a_{45} as the warm-started seeds (Supplementary Section S5), so the warm start does not bias stage (iii). A $\pm 4\,\mathrm{SD}$ box removes less than 10^{-4} of a Gaussian marginal, so narrowing shortens the tempering without truncating the posterior. If more than 20% of the posterior mass of an entry lies in the outer 5% of the box on one side, or this fraction grows by more than 0.10 from the previous stage, the bound is moved outwards and the stage is repeated; the boxes finally used are listed in Supplementary Table S4.

TMCMC on the GPU.. TMCMC [8, 14] passes a particle population through the tempered posteriors $p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta})^{\beta_m} p(\boldsymbol{\theta})$, $0 = \beta_0 < \cdots < \beta_K = 1$ (K tempering steps), choosing each increment so that the coefficient of variation of the importance weights stays at a target value [9]. After resampling, every particle performs n_{mut} Metropolis–Hastings steps with random-walk proposals scaled by the weighted particle covariance, alternating with differential-evolution proposals [15] that follow strongly correlated directions; proposals outside the box are rejected without solving the model. Each likelihood evaluation requires one solution of Eqs. (3)–(4), and a run requires

10^5 to 3×10^7 of them. We implement the implicit solver, including Jacobian assembly and linear solve, in JAX, compile it with the XLA (Accelerated Linear Algebra) compiler into a single kernel and evaluate all particles at once with `vmap`. On one GPU (NVIDIA RTX 4090) this is 170 times faster per tempering step than a 12-core CPU at matched settings (147 times end to end; Supplementary Table S3), which makes it affordable to repeat every stage with three seeds.

Acceptance criteria.. Every stage is run with three independent seeds and accepted only if (a) the stored log-likelihood of every particle agrees with a recomputation; (b) the tempering reaches $\beta = 1$; (c) each particle is moved at least twice per tempering step and at least five times per free parameter over the run; (d) the maximum log-likelihood, refined by a local optimisation (L-BFGS-B) from the best particles of each seed, differs by at most 1 across seeds, i.e. all seeds reach the same likelihood peak; and (e) every posterior median differs across seeds by at most half a posterior SD; an entry whose pooled posterior SD is at least 0.8 times that of a uniform distribution on the prior box of the stage is reported as not identified and excluded from this criterion, and for an entry with several posterior modes the criterion is applied within each mode, with the mode weights reported together with their spread across seeds. We call all other entries identified: the data constrain them beyond their prior box. Stages (ii)–(iv) that fail are repeated with more particles or mutation steps and not interpreted; the one exception, stage (iv) of DH, is discussed in Section 4.5. Stage (i) is exploratory: it sets the stage-(ii) box and gives the evidence estimates used for model comparison (Section 4).

4. Results and discussion

4.1. Fit and convergence

Stages (ii)–(iv) of CS and CH, stages (iii)–(iv) of DS and stages (ii)–(iii) of DH meet criteria (a)–(e) with three seeds each. Stage (ii) of DS fails (d), one seed having found a higher optimum (spread 1.7); the following stages, which start half of their particles from the prior, do not inherit this. Stage (iv) of DH, on the narrowed box, meets (a)–(d) but not (e): the seed medians of a_{13}, a_{22} and a_{24} differ by up to 0.79 posterior SD, so the three seeds settle in different modes. For DH the final stage is therefore stage (iii). None of the exploratory stage-(i) groups meets (c) (2.7–4.2 moves per

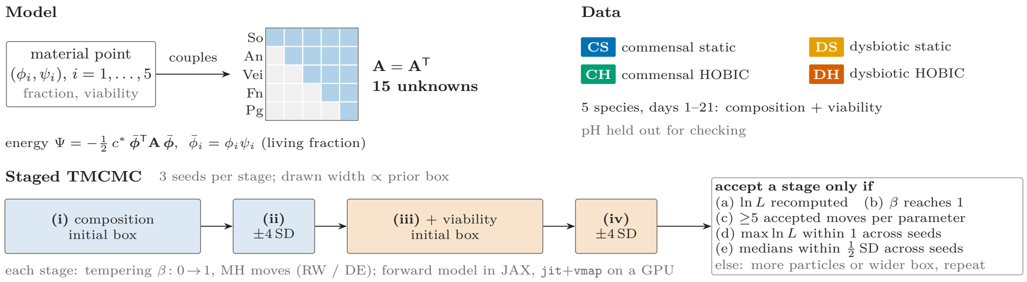
Model
material point
(ϕ_i, ψ_i), $i = 1, \ldots, 5$
fraction, viability
couples
So
An
Vei
Fn
Pg
$\mathbf{A} = \mathbf{A}^\mathsf{T}$
15 unknowns
energy $\Psi = -\frac{1}{2} c^* \bar{\boldsymbol{\phi}}^\mathsf{T} \mathbf{A} \bar{\boldsymbol{\phi}}$, $\bar{\phi}_i = \phi_i \psi_i$ (living fraction)
Data
CS commensal static
DS dysbiotic static
CH commensal HOBIC
DH dysbiotic HOBIC
5 species, days 1–21: composition + viability
pH held out for checking
Staged TMCMC
3 seeds per stage; drawn width ∝ prior box
(i) composition initial box
(ii) ±4 SD
(iii) + viability initial box
(iv) ±4 SD
accept a stage only if
(a) $\ln L$ recomputed (b) β reaches 1
(c) ≥5 accepted moves per parameter
(d) $\max \ln L$ within 1 across seeds
(e) medians within $\frac{1}{2}$ SD across seeds
else: more particles or wider box, repeat
each stage: tempering $\beta: 0 \to 1$, MH moves (RW / DE); forward model in JAX, jit+vmap on a GPU

Figure 1: Model and inference. A material point carries the volume fractions ϕ_i and viabilities ψ_i of five taxa, coupled through the symmetric interaction matrix $\mathbf{A}$ (15 unknowns). TMCMC with a GPU-compiled forward model infers $\mathbf{A}$ in four stages, from species composition to community viability; every stage is accepted only after the checks (a)–(e).

free parameter) (Supplementary Table S5). At the final stage the viable composition is reproduced with an RMSE of 0.060–0.066 in CH (MAP of each of the three seeds; total-count composition 0.076–0.082, live fraction 0.11–0.12), 0.090–0.093 in CS (total-count composition 0.039–0.041, live fraction 0.23; Section 4.5) 0.057 in DS (total-count composition 0.081, live fraction 0.09–0.10) and 0.073–0.076 in DH (total-count composition 0.155–0.159, live fraction 0.07–0.08) (Fig. 2). Accepting a stage often required more than one attempt (widened bounds in every condition, up to 20000 particles in DH; Supplementary Tables S4 and S5); without the criteria, several of these runs would have been read as converged.

4.2. What the data determine

Figure 3 and Supplementary Table S7 classify every entry of $\mathbf{A}$ at the final stage, and Supplementary Fig. S2 shows the marginal posteriors; the number of directions in which the likelihood itself is narrow at the MAP is 5 in CS, 6 in CH, 4 in DS and 7 in DH, against 5, 6, 9 and 8 identified entries (Supplementary Section S6); the counts depend on the threshold 0.8, the pattern described below does not (Supplementary Table S6). On synthetic data generated from a point of the CH posterior, the same pipeline reproduces this classification entry by entry (the six identified entries are identified again, the nine others are not), while only two of the six 90% intervals of the identified entries contain the true value (Supplementary Section S7). In the commensal conditions only a few entries are identified: five in CS (a_{11}, a_{22}, a_{33}, a_{13}, a_{23}) and six in CH (a_{11}, a_{12}, a_{22}, a_{33}, a_{13}, a_{23}), all among the early colonisers

So, An and Vei. Where both conditions determine the sign of an entry, it agrees: the self-interactions of So and Vei are negative (a_{11}, 90% intervals $[-7.0, -1.6]$ in CS and $[-1.4, -0.4]$ in CH; a_{33}, Section 4.5 and $[-5.8, -0.6]$) and the So–Vei coefficient is positive (a_{13}, $[+4.7, +11.3]$ and $[+0.3, +2.4]$), in line with the lactate cross-feeding between streptococci and *Veillonella* [16], although a_{13} is an effective coefficient and not a measurement of that mechanism. No entry involving Fn or Pg is identified in CH: their posteriors spread over nearly the whole box (posterior SD 0.94–1.00 times that of the uniform distribution), and none is determined even in sign. This is expected: Fn and Pg stay below 1% of the commensal communities from day 3 on, so the composition data carry little information on their interactions. In the dysbiotic conditions, where Fn and Pg reach 10–27% of the community, more entries are identified: nine in DS and eight in DH, now including interactions of the late colonisers. The Vei–Pg coefficient a_{35} is identified in both but differs between them: negative in the flow reactor (DH, $[-20.9, -0.6]$) and close to zero under static cultivation (DS, $[-0.34, +0.24]$); the An–Fn coefficient a_{24} is positive in DH ($[+1.3, +8.8]$) and not sign-determined in DS ($[-0.1, +2.9]$). A sign change between cultivation modes is a statement about the effective model under different nutrient and flow conditions, not about the pairwise mechanism itself. Comparisons between the commensal and the dysbiotic community are confounded by design: the two communities differ not only in their ecological state but also in the *Veillonella* species (*V. dispar* vs. *V. parvula*) and the Pg strain (DSM 20709 vs. W83) [2], so differences between their matrices combine state, strain physiology and cultivation; the DS–DH contrast above compares the same strains and is free of this confound, although the two conditions also differ in the viability weight ($\lambda_\psi = 2.0$ vs. 3.0) and in their prior boxes.

The classification is a result in its own right: it states which conclusions about the interaction network the data can support, and it is obtained from the same runs at no additional cost.

4.3. The Fn–Pg coefficient

The coefficient a_{45} links the bridging organism Fn to the pathogen Pg. Comparing the full model with a model in which $a_{45} = 0$, both inferred at stage (i) with 120 mutation steps so that both groups meet the acceptance criteria, gives $\ln B = -0.06$ (CS) and $+0.03$ (CH) (Supplementary Table S9): in the commensal conditions the data do not distinguish the two models, as expected for a coefficient between two taxa below 5%. In the dysbiotic conditions the data favour the full model, $\ln B = +7.5$ (DS)**[DS: provisional until the $a_{45} = 0$ group passes (d)]** and +2.1 (DH), by more than the seed spread of $\ln \hat{Z}$. Because the uniform prior penalises the full model by the log-width of the a_{45} box, $\ln B$ is read only within a condition: in DS, where the stage-(i) box is $[-15, 20]$, the penalty alone is $\ln 35 \approx 3.6$. In DH, the data favour a non-zero coefficient. With the composition alone (stage (i)), a_{45} in DH is positive in all seeds (medians +4.7 to +5.1, 90% intervals from $[+0.2, +5.9]$ to $[+2.0, +5.9]$) within the stage-(i) box, which allows only small negative values ($[-0.5, 6]$). Once the viability enters, on a box widened to $[-5, 20]$ in DH, the median of a_{45} stays positive in both dysbiotic conditions (+0.6 in DS, +3.1 in DH) but the 90% intervals include zero ($[-0.25, +2.1]$ and $[-2.4, +9.2]$; Supplementary Fig. S2): a_{45} is constrained but not sign-determined. Within the stage-(i) box, the composition data thus favour a positive Fn–Pg coefficient in DH, and the viability data neither confirm nor contradict it.

4.4. Checks against data not used in the inference

The pH measured under dynamic cultivation was not used in the inference (Fig. 4b,c). A linear regression of the measured pH on the measured species fractions ($R^2 = 0.88$ over the continuous records of CH and DH) was fitted once and applied, unchanged, to the posterior predictive composition, so the check tests the predicted composition, not a model of pH. In DH the pH predicted in this way follows the measured rise from about 6.4 to 7.0 ($R^2 = 0.71$, RMSE 0.10, 85% of the measured points inside the 90% band; the regression applied to the measured composition itself gives RMSE 0.12). In CH the measured pH stays near 6.3; the prediction matches this level (RMSE 0.08) but not its small variations ($R^2 < 0$, 44% inside the band). The calibrated model can also be used in silico (Fig. 4a). With all five taxa, the day-21/day-15 ratio of Pg in DH has a posterior median of 11–13 (one value per seed; 90% intervals up to $[0.6, 32]$; measured 14.9): the model reproduces the late increase, but the data do not determine its size (Section 4.5). Removing Fn lowers the ratio to 3.0 (90% interval $[1.0, 4.8]$, the same in all seeds), so the increase is reduced about fourfold but not abolished. Setting only $a_{45} = 0$ removes it in most samples (median 0.8–0.9 in all seeds), so the increase rests mainly on the Fn–Pg coefficient, partly offset by the other interactions of Fn; the predicted reduction can be tested directly by a co-culture without Fn. Under static cultivation the measured increase is smaller (ratio 2.5) and the model reproduces part of it (1.4–1.5); there, removing Fn brings the ratio to 1.0–1.1, whereas $a_{45} = 0$ leaves it unchanged (1.5), so the increase in DS is carried by the other interactions of Fn rather than by a_{45}.

4.5. Limitations

The likelihood treats the five species fractions as independent although they sum to one, so the posterior intervals are too narrow: in the recovery test on synthetic CH data, four of the six identified entries have the true value outside their 90% interval (Supplementary Section S7); the intervals of the identified entries are lower bounds on their uncertainty, whereas the classification into identified and not identified is reproduced. With both compositions and the live fraction in the likelihood, the three data sets are matched together in CH, but not in CS. In CS the compositions and the community live fraction cannot be matched at the same time: raising the weight of the live fraction from $\lambda_\psi = 1.5$ to 24 lowers its error from 0.23 to 0.05 but raises the error of the viable composition from 0.09 to 0.16 (deterministic optimisation on the measured data; Supplementary Section S14). In CS the data also leave the lower end of the self-interaction a_{33} open: the likelihood is flat for $a_{33} \lesssim -5$, the posterior fills the prior box down to its lower bound however far the box is widened, and a_{33} is therefore reported only as an upper bound ($a_{33} < -5.3$ with 95% posterior probability). Neither an initial state from the day-1 data nor a day-specific σ_ℓ changes this (Supplementary Section S14). In the model the viabilities relax within the first time steps to values set by the interaction terms, and the So-dominated community that the compositions require has a live fraction above 0.9, against the measured 0.65–0.77; without antimicrobial agents ($\alpha^* = 0$) the viability equation has no term that would keep part of a growing population dead. We report this tension rather than tune λ_ψ for CS. In DH the late increase of Pg is not determined by the data: the ratio of the Pg fraction on day 21 to day 15 is 9.6, 13.5 and 18.0 at the MAPs of the three stage-(iii) seeds (measured 14.9), whose log-likelihoods differ by less than 0.4, and the pooled posterior gives a median of 11–13 per seed with 90% intervals reaching from 0.6 to 32. On the narrowed box of stage (iv) the seeds settle in different modes (ratios 11.5, 13.7 and 18.9), which is why stage (iii) is the final stage for DH; treating these modes separately with more seeds is left to future work. The two composition channels constrain the viabilities ψ_i species by species; in CH the model follows the data for the dominant species So, in CS it keeps the main species almost fully viable (Supplementary Section S13). The interaction coefficients are effective quantities of this model and data set; they lump mechanisms that the model does not resolve, such as metabolite exchange and pH, and are not direct measurements of pairwise mechanisms. The calibration covers a 0D material point; the spatial structure of the biofilm and its mechanics are not part of the inference. The calibrated dynamics are, however, the input that continuum growth models of biofilms require [1], so that differences in growth-induced stress follow from calibrated interactions rather than assumed growth weights; since the likelihood uses fractions only, the absolute biomass is a model prediction whose posterior spread has to be propagated into such a coupling. At a material point constrained in all directions, growth driven by the living volume separates the conditions, whereas growth driven by the total volume, which the data do not constrain, does not (Supplementary Section S16).

5. Conclusions

The interaction matrix of a variational multiphase biofilm model was calibrated against time courses of a five-species oral community in four conditions, with a GPU-compiled forward model, a staged likelihood and explicit acceptance criteria for every stage. Three points carry over to other inverse characterisations of living materials. First, convergence has to be checked, not assumed: posterior mass at the prior bounds was found in every condition and disagreement between seeds in the dysbiotic ones, and both were addressed only by widening bounds and adding particles. Second, identifiability is a result: with 20 independent composition values and five viability values for 15 unknowns the data determine only some interactions, and stating which ones limits the biological conclusions that can be drawn from the model. Third, a misfit can be informative: in the commensal static condition the composition and the viability data cannot be matched at once, which may point to a process the model lacks, or to a mismatch between the live/dead stain and the model viability, rather than to a failed calibration. In the dysbiotic conditions more interactions are identified, and the Fn–Pg coefficient, which the composition data favour as positive in DH within the stage-(i) box, is constrained but not sign-determined by the full data set.

CRediT authorship contribution statement

[draft, to be confirmed by all co-authors]

Keisuke Nishioka: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft.
Felix Klempt: Methodology, Software, Validation, Writing – review & editing.
Hendrik Geisler: Validation, Writing – review & editing.
Rumjhum Mukherjee: Validation, Writing – review & editing.
Nils Heine: Resources, Data curation, Writing – review & editing.
Katharina Doll-Nikutta: Resources, Writing – review & editing.
Meike Stiesch: Resources, Funding acquisition, Writing – review & editing.
Szymon P. Szafrański: Validation, Writing – review & editing.
Meisam Soleimani: Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.
Philipp Junker: Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

Code, run configurations, convergence records and posterior samples are archived at Zenodo (doi:10.5281/zenodo.23320423). The experimental data are those of Heine et al. [2] and are available from those authors; they are not redistributed with the archive.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to improve the language and readability of the manuscript and to assist with programming the analysis and plotting scripts. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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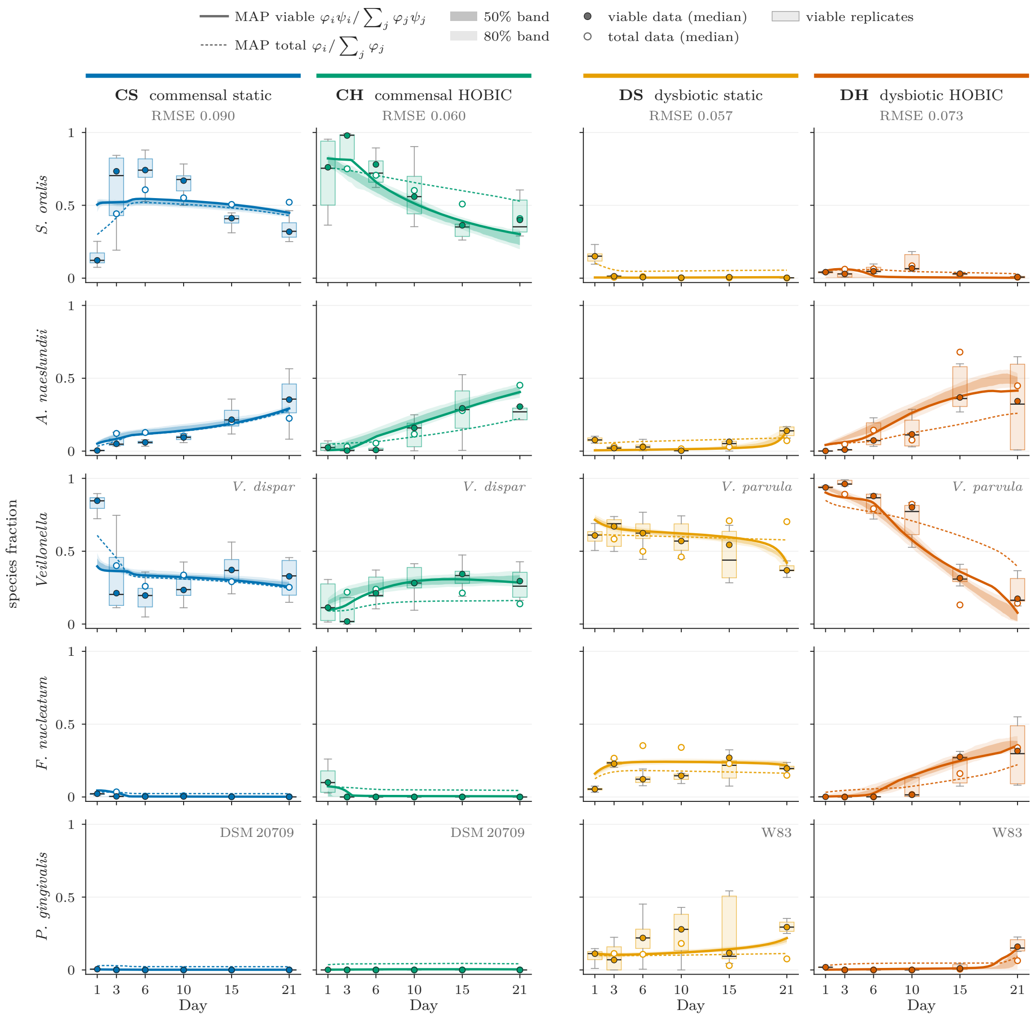

Figure 2: Species fractions over 21 days in the four conditions (columns) for the five taxa (rows). Solid lines: MAP of the final stage (stage (iv); stage (iii) in DH), living fractions $\phi_i\psi_i/\sum_j \phi_j\psi_j$, with the viable-cell data (filled circles: medians; boxes and whiskers: interquartile range and range of the measured replicates); bands: central 50% and 80% intervals of the living fractions over the posterior samples, pooled over three seeds. Dashed lines: MAP volume fractions $\phi_i/\sum_j \phi_j$, with the total-count data (open circles: medians). The RMSE of the viable composition at the MAP is given above each column. The *Veillonella* species and the Pg strain differ between the commensal (CS, CH) and the dysbiotic (DS, DH) community (labels in the panels).

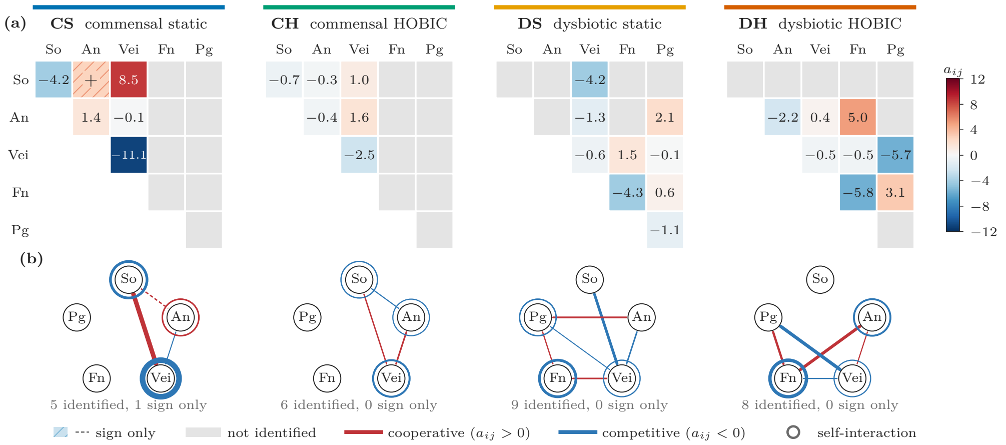

Figure 3: What the data determine, at the final stage with three seeds pooled. (a) Upper triangle of $\mathbf{A}$ in the four conditions. Coloured with value: identified, i.e. constrained by the data beyond the prior box (posterior SD below 0.8 times that of a uniform distribution on the prior box of the final stage, Supplementary Table S4); colour and number give the posterior median; hatched: only the sign is determined (90% interval excludes zero); grey: not identified. (b) The same entries as a network: red, $a_{ij} > 0$; blue, $a_{ij} < 0$; line width proportional to $|\text{median } a_{ij}|$; dashed, sign only; rings, identified self-interactions a_{ii}. Entries that are not identified are not drawn.

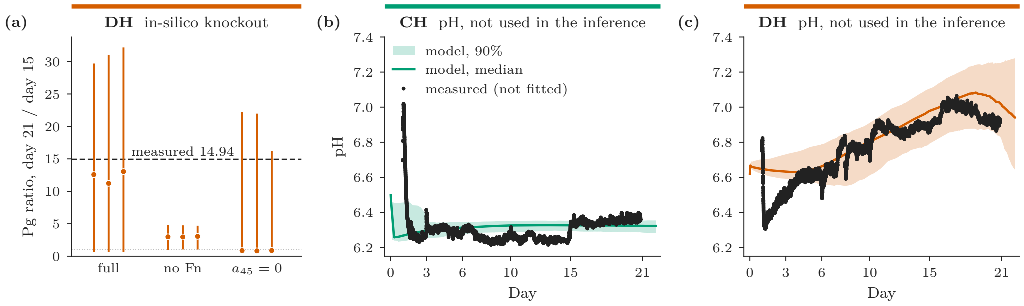

Figure 4: Checks against data and predictions not used in the inference. (a) DH: ratio of the Pg fraction on day 21 to day 15 for the calibrated model, with Fn removed, and with $a_{45} = 0$ (posterior median and 90% interval, one bar per seed; dashed: measured). (b,c) pH predicted from the posterior composition through a regression fitted once on the measured pH and composition (band: central 90% interval), against the measured pH (dots: continuous record, mean of the reactors). DH: final stage (iii).